

M002 Week 5 — Mini Cube Puzzle

Topic: Mini Cube Puzzle · **Course:** Maze Madness · **Level:** Intermediate · **Time:** about 35 minutes

Finish the mini cube. Add a conditional, fill every layer, then break the top and climb out.

1 · Predict

Write what happens.

```
repeat 5 times:
  place block
  move forward
```

What shape is left? How long?


```
if on the edge OR on the top layer:
  place block
otherwise:
  leave empty
```

Only one part must be true. Which spots get a block?

2 · Plain to Code

Match the plain words to the API.

```
move up one block
```

Word bank: `agent.move(UP, 1)` · `agent.move(FORWARD, 1)` · `agent.turn(LEFT_TURN)`

Write the line:

3 · Trace 🦶

```
agent.place(BLOCK)
agent.move(FORWARD, 1)
agent.place(BLOCK)
agent.move(UP, 1)
```

How many blocks placed? Where does the agent end — on the floor or one up?

4 · Spot the Bug 🐛

Bug A — The agent should fill the layer, **then** move up. It moves up too soon.

```
agent.place(BLOCK)
agent.move(UP, 1)
agent.place(BLOCK)
```

Fix it:

Bug B — To climb out, make the hole **first**, then move up. Fix the order.

```
agent.move(UP, 1)
agent.detect(BLOCK, UP)
```

Hint: break the block above first.

Fix it:

5 · Show Your Work 📷 🎥

Open this week's world. Find the **mini cube starter** — part is built.

1. Keep the starter going.
2. Add one conditional that uses **OR**.
3. Fill every layer.
4. Break a hole in the top and climb out.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built ____.

I built it using this code: ____.

In this code I used ____.

The hardest part was ____.

That part was hard because ____.

The most fun part was ____.

Something new I learned was ____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.